Ekaterina Landgren

Contact Information

Department of Environmental Social Sciences
Stanford Doerr School of Sustainability
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Research interests

Complex social systems: environmental social science, computational social science, opinion dynamics
Mathematics of climate: conceptual climate models, exoplanetary atmosphere dynamics

Education

Cornell University **2022**
Ph.D., Applied Mathematics
"Models of Varying Complexity from Voter Networks to Extrasolar Planets"
Advisor: Steven H. Strogatz

Cornell University **2020**
M.S., Applied Mathematics

Brown University **2017**
B.S., Applied Mathematics, B.A., Philosophy
Cum Laude, Phi Beta Kappa, Sigma Xi
"Modeling Evacuation Dynamics in a Crowded Room"
Advisor: Bjorn Sandstede

Professional Experience

Stanford University **September 2025–present**
Stanford Doerr School of Sustainability
Dean's Sustainability Leaders Postdoctoral Fellow

University of Colorado Boulder
Cooperative Institute for Research in Environmental Sciences
Postdoctoral Researcher **2025**
Visiting Postdoctoral Fellow **2023–2024**

Awards and Fellowships

Best Postdoc Presentation 2026
Awarded at the Sustainability Data Science Conference held at Stanford University.

Rising Star in Computational and Data Sciences 2025
Oden Institute, University of Texas at Austin

Best Oral Presentation by an Early-Career Researcher, Honorable Mention 2024
Awarded at International School and Conference on Network Science (NetSci 2024).

SIAM Science Policy Fellowship 2024

Awarded annually to 5 early-career mathematicians to gain in-depth knowledge of science policy.

Collaborate@ICERM 2024

Awarded to a team of 5 mathematicians to spend a week collaborating on the project "Modeling and Analysis of Candidate Momentum in U.S. Primary Elections."

Zonta International Amelia Earhart Fellowship 2021

Awarded annually to up to 35 women around the globe pursuing a PhD in space sciences.

SIAM Student Chapter Certificate of Recognition 2021

Awarded for outstanding service and contributions to the SIAM student chapter.

Undergraduate Research and Teaching Award 2015, 2016

Awarded to Brown students collaborating with faculty on research projects.

Mathematical Contest in Modeling, Honorable Mention 2016

In an undergraduate team, created, analyzed, and wrote a report on a model of fluid dynamics.

Brown Mathematical Contest for Modeling, Outstanding Winner 2015

In an undergraduate team, created, analyzed, and wrote a report on a model of Hantavirus spread.

Travel awards

yrCSS Bridge Grant	2025
Dynamics Days 2025 Travel Award	2025
AWM Travel Grant	2024
Postdoctoral Association of Colorado Travel Award	2024
yrCSS Scholarship for Events on Complex Systems	2024
SIAM Early Career Travel Award	2024
CIRES Early Career Travel Award	2024
Dynamics Days 2024 Travel Award	2024
SIAM Student Travel Award	2019

Submitted Manuscripts

*Equal contribution indicated by **

1. Consensus and fragmentation in academic publication preferences
   Ian Van Buskirk, Marilena Hohmann*, **Ekaterina Landgren*,** Johan Ugander, Aaron Clauset, Daniel Larremore
   arXiv preprint: https://arxiv.org/abs/2603.00807
2. Can homophily explain public underestimation of climate policy support?
   Ekaterina Landgren, Shriya Nagpal, Joshua Garland, Yaw Acquah, Matthew G. Burgess
   arXiv preprint: https://arxiv.org/abs/2606.15106

Peer-Reviewed Publications

1. Optimal ambition in business, politics, and life
   Ekaterina Landgren, Ryan Langendorf, Matthew Burgess
   Physical Review E 113, 054317 (2026)*. Selected as Editors' Suggestion.*
   DOI: 10.1103/dfw8-vhjk
   Preprint: https://arxiv.org/abs/2502.10500

Alphabetical author order indicated by ✦

2. U.S. television news coverage of climate change policy is aggregately balanced but polarized
   Ekaterina Landgren, Jeremiah Osborne-Gowey, Joshua Garland, Maxwell Boykoff, Matthew Burgess
   Environmental Research Communications 8 051016 (2026). DOI: 10.1088/2515-7620/ae6b10

3. Visualizing epidemiological models for policy: design principles for effective communication
   Liza Hadley, Nick Holliman, Kai Xu, Edyta Bogucka, Xinhuan Shu, Daniel Archambault, **Ekaterina Landgren,** Ellen M. DeGennaro, Stephen M. Kissler.
   Frontiers in Public Health 14:1798154 (2026). DOI: 10.1103/dfw8-vhjk
   Preprint: https://doi.org/10.17605/OSF.IO/C93HK

4. A Shallow-water Model Exploration of Atmospheric Circulation on Sub-Neptunes: Effects of Radiative Forcing and Rotation Period
   Ekaterina Landgren, Alice Nadeau, Nikole Lewis, Tiffany Kataria, Peter Hitchcock
   *Planetary Science Journa*l, 4(6), 106. (2023). DOI: 10.3847/PSJ/acd551

5. SWAMPE: A Shallow-Water Atmospheric Model in Python for Exoplanets
   Ekaterina Landgren, Alice Nadeau
   Journal of Open Source Software 7 (80), 4872 (2022). DOI: 10.21105/joss.04872

6. Comparison of Two Analytic Energy Balance Models Shows Stable Partial Ice Cover Possible for Any Obliquity
   Ekaterina Landgren, Alice Nadeau
   Planetary Science Journal 3.79 (2022). DOI: 10.3847/PSJ/ac603d

7. How a minority can win: Unrepresentative outcomes in a simple model of voter turnout
   Ekaterina Landgren, Jonas L. Juul, Steven H. Strogatz
   Physical Review E 104.5 (2021): 054307. DOI: 10.1103/PhysRevE.104.054307
   Preprint

8. Fractal Behavior of the Fibonomial Triangle Modulo Prime p, Where the Rank of Apparition of p is p + 1
   ✦ Michael DeBellevue, **Ekaterina** Kryuchkova **(Landgren)**
   Fibonacci Quarterly 56 (2018): 113-120.

Other writing

1. Modeling Social Systems: Transparency, Reproducibility, and Responsibility
   ✦Maximino Aldana, Heather Zinn Brooks,, Phil Chodrow, Fillipe Georgiou, Joseph Johnson, Krešimir Josić, Zachary Kilpatrick, **Ekaterina Landgren,** Andrew Nugent, Maurizio Porfiri, Nancy Rodriguez, Pablo Suárez-Serrato, Roni Barak Ventura, David White, Alexander Wiedemann, Sam Zhang.
   SIAM News 59.2 (2026). DOI: 10.48550/arXiv.2508.18542

Presentations

Invited presentations

1. *Perceived climate polarization from networks to newsrooms* — March 2026
   Lund University COSHE Seminar
2. *Publication preferences of academics* — October 2025
   University of Virginia, Seminar, Ahn lab
3. *Perceived climate polarization from networks to newsrooms* — July 2025
   IT University of Copenhagen Seminar, Copenhagen, Denmark
4. *Disentangling climate polarization* — June 2025
   Mathematics of Climate Research Network Colloquium
5. *Disentangling climate polarization* — June 2025
   BIRS Workshop: Collective Social Phenomena
   Casa Matemática Oaxaca, Oaxaca, Mexico
6. *Unpacking Climate Polarization Across News and Social Media* — May 2025
   SIAM Conference on Applied Dynamical Systems, Denver, CO
7. *Disentangling climate polarization* — April 2025
   Rising Stars in Computational and Data Sciences, Oden Institute, Austin, TX
8. *Disentangling climate polarization on Reddit* — April 2025
   CIRES Early-Career Assembly Seminar, Boulder, CO
9. *Modeling misperception of public support for climate policy* — November 2024
   University of Minnesota, School of Mathematics, Minneapolis, MN
10. *Exoplanets in 1D and 2D: from ice cover to atmospheric circulation* — November 2024
    University of Minnesota, Mathematics of Climate Seminar, Minneapolis, MN
11. *Modeling misperception of public support for climate policy* — October 2024
    Stanford University, Climate Cognition and Sustainability and Social Change Lab
12. *Modeling misperception of public support for climate policy* — April 2024
    National Ecological Observatory Network (NEON) Science Seminar
13. *Modeling misperception of public support for climate policy* — March 2024
    University of Vermont, Complex Systems Center, Burlington, VT
14. *Modeling misperception of public support for climate policy* — February 2024
    University of Colorado Boulder, Dynamical Systems Seminar, Boulder, CO
15. *Modeling misperception of public support for climate policy* — February 2024
    University of Minnesota, Mathematics of Climate Seminar
16. *Modeling misperception of public support for climate policy* — December 2023
    University of Colorado Boulder, Mathematical Biology Seminar
17. *A Shallow Water Model of Atmospheric Circulation on Sub-Neptunes* — November 2023
    Max Planck Institute for Astronomy, Exocoffee
18. *Misperception of public support for climate policy: A Networks Perspective* — October 2023
    University of Cambridge Centre for Climate Repair, Cambridge, UK
19. *Beyond Echo Chambers: Misperception of Public Support for Climate Policy* — September 2023
    Brown University, LCDS Seminar, Providence, RI

20. *Modeling Misperception of Public Support for Climate Policy* May 2023
SIAM Conference on Applied Dynamical Systems, Portland, OR
21. *A Shallow-Water Model Exploration of Atmospheric Circulation on Sub-Neptunes* April 2023
Southwest Research Institute, Boulder, CO
22. *How Can Minority Win?* February 2023
University of Colorado Boulder, Seminar, Clauset & Larremore lab group
23. *Introduction to Research* February 2022
Cornell Chapter of Association for Women in Mathematics, Ithaca, NY
24. *Effects of Network Structure on Undemocratic Outcomes* August 2021
Clarkson University, Graduate Student Seminar
25. *Effects of Network Structure on Undemocratic Outcomes* May 2021
SIAM Conference on Applied Dynamical Systems
26. *Noisy El Niño: A Case Study of Conceptual Climate Models* March 2021
Mount Holyoke College, Math and Statistics Tea
27. *When Can Minority Win? A Simple Model of Voter Turnout* February 2021
Women in Network Science Seminar, University of Washington
28. *Snowball Planets: Effects of Obliquity, Albedo, and Heat Transport on Ice Cover* October 2020
Jet Propulsion Laboratory, Exoplanet Journal Club

Contributed presentations

- *Climate attribution in local news coverage of natural hazards* April 2026
  Sustainability Data Science Conference, Stanford University
- *The publication preferences of academics* July 2025
  International Conference on Computational Social Science, Norrköping, Sweden
- *Optimal ambition in business, politics, and life* July 2025
  International Conference on Computational Social Science, Norrköping, Sweden
- *Modeling misperception of public support for climate policy* June 2024
  International School and Conference on Network Science, Québec City, Canada
- *Modeling misperception of public support for climate policy* June 2024
  SIAM Conference on Mathematics of Planet Earth, Portland, OR
- *Modeling misperception of public support for climate policy* April 2024
  Network Inequality Seminar, Complexity Science Hub
- *Modeling misperception of public support for climate policy* January 2024
  Dynamics Days 2024. University of California, Davis
- *Climate policy is more popular than most people think* October 2023
  Social and Environmental Futures Workshop, University of Colorado Boulder
- *How can minority win?* August 2022
  Contagion on Complex Social Systems Workshop, University of Colorado Boulder
- *Introducing SWAMP-E: Shallow Water Atmosphere Model in Python for Exoplanets* May 2021
  Emerging Researchers in Exoplanet Science Conference

Poster presentations

- *Optimal Ambition in Business, Politics, and Life* January 2025
  Dynamics Days 2025. Denver, CO

- *Modeling misperception of public support for climate policy* — July 2024
  International Conference on Computational Social Science, Philadelphia, PA
- *Climate policy is more popular than you think!* — March 2024
  STEM Poster Day at the Colorado State Capitol
  Project Bridge, University of Colorado Anschutz
- *Exploring the Interaction of Rotation Rate and Stellar Irradiation on Synchronously Rotating Sub-Neptunes* — December 2022
  American Geophysical Union Fall Meeting. Chicago, IL
- *Introducing SWAMP-E: Shallow-Water Atmospheric Model in Python for Exoplanets*
  American Geophysical Union Fall Meeting — December 2021
- *Introducing SWAMP-E: Shallow-Water Atmospheric Model in Python for Exoplanets*
  Emerging Researchers in Exoplanet Science Conference — May 2021

Student Mentorship

Daniel Verdi — 2026
Ph.D. Student in Education at Stanford University
Mentored jointly with Sara Constantino.
Project title: "Climate change in political speech"

Isha Sinha — 2026
M.Sc. Student in Computer Science at Stanford University
Mentored jointly with Sara Constantino.
Project title: "Climate change in political speech"

Ian van Buskirk — 2024–2025
Ph.D. Student in Computer Science at the University of Colorado Boulder
Mentored jointly with Dan Larremore.
Project title: "Publication preferences of academics"

News Transcript Annotation Team — 2023–2024
Team of undergraduate and master's students at the University of Colorado Boulder
Callie Blaseg, Kai Elznic, Lucas Gauthier, Katherine Konecny, Piper Nilsen, Vraj Patel, Trishala Thakur
Led twice-weekly intercoder reliability meetings and supervised news transcript annotations.
Mentored jointly with Jeremiah Osborne-Gowey.
Project title: "Misperception of public support for climate policy"

Thomas Mitchell — 2022
Undergraduate Student in Astronomy at Cornell University
Mentored jointly with Nikole Lewis.
Project title: "Energy Balance Model for HAT-P-2b"

Anna Asch — 2021
Undergraduate Student in Mathematics at Cornell University
Mentored jointly with Shriya Nagpal and Alice Nadeau.
Project title: "Wind farm layout optimization"

Anna Asch — 2020
Undergraduate Student in Mathematics at Cornell University

Directed Reading Program
Project title: "Mathematics and Climate"

Anushka Naranayan 2020
Undergraduate Student in Mathematics at Cornell University
Mentored jointly with Alice Nadeau.
Project title: "Applying the Budyko Model to Martian Obliquity"

Teaching Experience

MIT Educational Studies Program
Instructor
M14095: Mathematical Models and How to Build One, *Online* Summer 2020
Designed and taught a six-session class in mathematical modeling for high school students.

Cornell University
Teaching Assistant
MATH 4210: Nonlinear Dynamics and Chaos Spring 2020
MATH 3610: Mathematical Modeling Fall 2019
MATH 2930: Differential Equations for Engineers Spring 2019

Brown University
Teaching Assistant
APMA 1650: Statistical Inference I Fall 2015, Spring 2017

Industry experience

IMA Math-to-Industry Bootcamp III Summer 2018
Six-week coding and research program. Minneapolis, MN
Hewlett-Packard Customer Operations Summer 2014
Summer intern. Moscow, Russia

Service and Leadership

Conference Session Organizer
- *Human behavior in climate opinion, policy, and environmental systems at SIAM DS25* May 2025
  Co-organizer
- *AMS Special Session on Complex Social Systems at JMM* January 2024
  Co-organizer
- *Dynamics of Influence and Representation in Social Systems at SIAM DS21* May 2021
  Co-organizer

Stanford University
- *Menlo School Catalyst Leadership Program* March 2026
  Guest speaker on career pathways in environmental social science
- *Women in Network Science Society*

Treasurer 2025–present
Communications team: publish a monthly newsletter. 2023–present

University of Colorado Boulder

- *CIRES Cross-Discipline and Community Building Mentoring Workshop* April 2025
  Panelist
- *Postdoctoral Mentoring Program* 2024
  Mentor
- *Mathematics of Climate Research Network Mentoring Program* 2024
  Mentor
- *Kent Denver School Gender Advancements in STEM Career Panel* January 2023
  Panelist

Cornell University

- *Expanding Your Horizons Conference* 2021
  Logistics chair, organized a campus-wide STEM outreach event for 500 middle-school girls.
- *Write a Researcher* 2021
  Corresponded with a high school student about mathematics research.
- *Center for Applied Mathematics First-Year Mentoring Program* 2019, 2021
  Mentored a first-year PhD student.
- *SIAM Graduate Student Chapter* 2018–2021
  President. Organized SIAM-sponsored events for student chapter members.
- *Center for Applied Math Anti-Racism Reading Group* 2020
  Co-organizer. Moderated a biweekly graduate student discussion focusing on DEI topics.
- *ZigZag Mentorship Program* 2017, 2019
  Mentored undergraduate students on course selection and career development.

Brown University

- *Applied Mathematics Department Undergraduate Group* 2015, 2016
  President. Organized events for undergraduates interested in applied mathematics.
- *Technology House* 2016
  President. Led a sixty-person, communal living group for students interested in STEM topics.
- *New Scientist Program* 2015
  Mentored and advised a first generation college student.

Reviewer for

Journal of Open Source Software, npj Complexity, Scientific Reports, Europhysics Letters, Physica D: Nonlinear Phenomena, IC2S2 2025, IC2S2 2026, Climatic Change, Environmental Research: Climate

Other Professional Activities

Workshops attended

- *Robust Institutional Design in Expert Decision Maker Systems,* Santa Fe Institute, NM April 2026
- *Complex Networks Winter Workshop,* Québec City, Canada December 2025
- *Social and Environmental Futures Workshop,* Boulder, CO October 2023
- *Mathematics Research Communities: Complex Social Systems,* Buffalo, NY June 2023

- *Contagion on Complex Social Systems,* Boulder, CO — August 2022
- *Science Communication Workshop,* Ithaca, NY — October 2021
- *Code/Astro: A Software Engineering Workshop for Astronomy,* Online — June 2021

Pedagogical development

- *Research Mentor Training* — Spring 2022
  Center for the Integration of Research, Teaching and Learning Network
  Completed a six-week online course to improve research mentoring skills

Membership in professional organizations

- Society for Industrial and Applied Mathematics
- American Mathematical Society
- Network Science Society
- Mathematics of Climate Research Network
- Women in Network Science Society

Media features

- "How Ambitious Should You Be? There's a Sweet Spot" *Time* (link) — June 2026
- "A Model for Ambition" *Physics Magazine* Focus article (link) — May 2026
- "Don't shoot for the moon: aiming for 'above average' is key to success, maths suggests" *The Guardian* (link) — May 2026
- "Aim high but don't shoot for the moon, mathematicians advise" *New Scientist* (link) — May 2026
- Featured in "Advancing Sustainability Through Data Science: Conference Highlights" in *Stanford Data Science News* (link) — May 2026
- Featured in "Postdoctoral fellows build research and community connections" in Stanford Doerr School of Sustainability news (link) — April 2026
- Featured in "2024 SIAM Science Policy Fellows" in SIAM news (link) — March 2024
- SIAM DS23 presentation featured in SIAM News Blog (link) — May 2023
- Amelia Earhart Fellowship Spotlight in the Cornell Chronicle (link) — May 2021